



Bayesian identification of oil spill source parameters from image contours

Samah El Mohtar^a, Boujemaa Ait-El-Fquih^a, Omar Knio^{a,b}, Issam Lakkis^c, Ibrahim Hoteit^{a,*}

^a King Abdullah University of Science and Technology, Thuwal 23955, Saudi Arabia

^b Duke University, Durham, NC 27708, USA

^c Maroun Semaan Faculty of Engineering and Architecture, American University of Beirut, Beirut, Lebanon

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ABSTRACT

Oil spills at sea pose a serious threat to coastal environments. Identifying oil pollution sources could help to investigate unreported spills, and satellite imagery can be an effective tool for this purpose. We present a Bayesian approach to estimate the source parameters of a spill from contours of oil slicks detected by remotely sensed images. Five parameters of interest are estimated: the 2D coordinates of the source of release, the time and duration of the spill, and the quantity of oil released. Two synthetic experiments of a spill released from a fixed point source are investigated, where a contour is fully observed in the first case, while two contours are partially observed at two different times in the second. In both experiments, the proposed method is able to provide good estimates of the parameters along with a level of confidence reflected by the uncertainties within.

1. Introduction

Seas around the world are experiencing a growth in maritime traffic. In fact, ship traffic has witnessed a fourfold increase between 1992 and 2012 (Tournadre, 2014). As such, oil discharge from ships at sea is becoming more frequent. Offshore platforms and pipelines are other sources contributing to the annual release of petroleum to the sea (National Research Council, 2003). Whether accidental or deliberate (through bilge dumping, for example), spill events pose a great threat to the coastal environment, and may cause considerable economic losses to marine activities disrupted by the pollution. To control and limit spills, it is essential to identify pollution sources, and satellite imagery can be an effective tool for this purpose. A similar framework would be also useful to determine the source of hydrocarbon marine seepage, from resulting surface slicks (Mityagina and Lavrova, 2016), for oil and gas exploration, and to backtrack other marine pollutants.

Various methods have been proposed to model the dispersion of pollutants (or tracers in general) in the ocean, following both Eulerian (e.g., Mouchet et al., 2016) and Lagrangian (e.g., Villa et al., 2015) approaches. Lagrangian particle tracking (LPT) methods are among the most commonly used, in which tracers are modeled by Lagrangian particles, whose trajectories are determined by a velocity field represented on an Eulerian grid (Van Sebille et al., 2018). Depending on the application, different modules can be built on top of LPT models to simulate a specific process, such as weathering (e.g., evaporation,

emulsification, etc.) in oil spill models (Spaulding, 2017), leeway drift in search and rescue applications (Breivik et al., 2013), and larval reaction in transport models of plankton (Batchelder et al., 2002), among others. LPT models require as input a set of parameters that characterize the release conditions of a polluting event, including the location of the source, the time and duration of the spill and the quantity of pollutant released. Given these source parameters that we henceforth group in a vector θ , the state and final location of the Lagrangian particles can be predicted by solving a forward problem that models the dispersion of a spill. In this work, we are interested in the solution of the inverse problem, that is the estimation of the source parameters of the spill from observations of oil pollution.

Backward tracking, or reverse-time trajectory modeling, has been proposed to identify the source of pollutants. In this approach, particles are released from a location where pollution is observed (final location from a forward-modeling point of view) and advected within a time-reversed flow field. Diffusion is often enabled in the backward mode and particles are treated as samples of different realizations of the source location (Batchelder, 2006). The results are presented in the form of probability or likelihood maps of particular locations being sources. Backward tracking has been employed in oil spill contingency planning to generate probability maps of locations, from which oil might reach high-valued coastal regions within a certain time (e.g., Torgimson, 1981; Galt and Payton, 1983; Ciappa and Costabile, 2014). The method has been also used to identify sources of plankton (e.g., Batchelder,

* Corresponding author.

E-mail address: ibrahim.hoteit@kaust.edu.sa (I. Hoteit).

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2006), backtrack satellite-detected oil slicks (e.g., Zodiatis et al., 2012), locate the origin of tar balls (e.g., Suneel et al., 2016), identify the sources of ocean litter (e.g., Ko et al., 2018) and determine the site of an airplane crash (e.g., Eichhorn and Haertel, 2016), among other applications. Some attempts have been made to enhance the estimation of the source location within the backward model. Isobe et al. (2009) proposed the two-way particle tracking method to reduce the number of source candidates. Thygesen (2011) presented a Bayesian approach to time reversal in stochastic particle tracking, in which a correction is added to the flow in the backward mode, to account for prior information about the source. El Mohtar et al. (2018) proposed a new method for backward tracking in stochastic flow fields generated within an ensemble data assimilation framework. Their method allows taking the uncertainties in the flow field into account, while controlling the exponential growth in the number of particles that arises from describing the behavior of the velocity over time as a set of possible combinations of the different realizations of the flow field.

Despite being straightforward to apply, backward tracking has some limitations. The method relies on observations of the location of pollutants at a given time, but cannot incorporate other available observations. As Thygesen (2011) pointed out, an issue that deserves development is the simultaneous backtracking of multiple particles and continuous fields. Another important limitation is that most of the processes that the particles undergo are time-irreversible, and thus, modules developed on top of forward trajectory models, which simulate such processes (e.g., oil weathering processes), cannot be included in the backward mode (Thygesen, 2011; Breivik et al., 2012). The method is also limited to the estimation of only the source and time of release. A more general method that relies on forward LPT models along with application-specific modules to estimate source parameters would be more appealing. Attempts have been made to use the forward model and take advantage of already-existing modules developed specifically for the application at hand. Banas et al. (2009) employed a forward LPT model that simulates larval behavior to determine crab larval retention in a bay. The end locations of the larvae were mapped to their release locations via lookup tables obtained from the forward model runs. Breivik et al. (2012) used an iterative seeding-and-selection algorithm with a (forward) trajectory model to determine the origin of a leeway-drifting object (e.g., a lifeboat). The iterative process allows retaining only the particles that fall within a certain time-space distance of the observation. Gautama et al. (2016) solved a minimization problem of the level-set distance between observed and modeled oil contours, using a forward trajectory model, to estimate the source parameters of an oil spill.

In environmental problems where observations are available in the form of concentration measurements of a contaminant (or any other measured quantity), Bayesian inference and Markov chain Monte Carlo (MCMC) techniques have been used to solve the inverse problem of source parameters' estimation. For instance, Delle Monache et al. (2008) employed a Bayesian approach along with MCMC sampling to estimate the source location and emission rate of a polluting event in the atmosphere on a continental scale. Ait-El-Fquih et al. (2020) estimated five source parameters in a groundwater pollution problem, using MCMC and a deterministic variational Bayesian approach. Unlike backward tracking, a Bayesian approach allows the estimation of different parameters, other than the source and time of release, along with parameters of the measurement error's probability distribution. Moreover, reversing time is not needed since the forward model (or its surrogate (e.g., Marzouk et al., 2007; Marzouk and Najm, 2009; Srarj et al., 2013; Giraldi et al., 2017; Srarj et al., 2017; Navarro et al., 2018)) is used within MCMC, which means any application-specific model can be employed without any modification. Prior knowledge about the parameters of interest can be further taken into account within the Bayesian framework, in the form of a probability distribution of those parameters. Finally, the method allows quantifying the uncertainties in the estimated parameters, which reflects a level of confidence in the final

solution. Given these advantages, we propose in this work a Bayesian approach for the estimation of oil spill source parameters, from observations of oil pollution.

Unlike observations in other environmental problems where pollution is measured by sensors at specific locations, marine pollution is more effectively monitored from space, in the form of binary images, or contours, that delineate the pollutant at the sea surface (Garcia-Pineda et al., 2009). Those contours are obtained by segmentation of grayscale images captured by satellite-based synthetic aperture radar (SAR) sensors. SAR sensors, however, are prone to speckle noise, which degrades the quality of the captured images, leading to an undesired granular aspect and a statistical distribution with large standard deviation, even for a homogeneous area (Brekke and Solberg, 2005). As such, the observed oil contours (or binary images) are affected by this measurement error, which introduces uncertainties into the observations. Moreover, depending on their field of view, available SAR images do not always capture the whole area of a spill, which means that in some cases parts of the contours are missed, and therefore, one has to rely on the observed parts and deal with the problem of missing data. Nonetheless, observations at different times may be available. To compare model simulations with observations, Lagrangian particles are sometimes overlaid over images of oil slicks (e.g., Cheng et al., 2011; Xu et al., 2013). Gautama et al. (2016) compared observed contours to modeled contours using a level-set method. They defined the modeled contours as the 85% highest-density contours (Haselsteiner et al., 2017) of the particles' density distribution obtained by kernel density estimation. In this study, we follow a similar approach to that of Gautama et al. (2016), however, we define modeled contours as contour lines of computed concentration fields, for a given concentration threshold. Given the oil density, this threshold is related to the minimum thickness of oil that can be detected by SAR sensors, at the edge of an observed oil slick (Fingas and Brown, 2018), which is also an uncertain parameter. For comparison, we employ a non-local metric (the global Hausdorff distance (Zhang and Leeuwenburgh, 2017)) to measure the dissimilarity between the observed and modeled contours.

The difference in the type of observations (binary images or contours as opposed to quantities such as concentrations) between marine pollution problems and classical Bayesian estimation problems calls for the need to develop a new (statistical) observation error model, and a likelihood function based on that model. Another contribution of this work is the introduction of such an observation error model (i.e., a likelihood) in which observations are image contours and not measured quantities. We propose a likelihood function such that maximizing it amounts to minimizing the proposed dissimilarity measure between modeled and observed contours. The likelihood is then used in a Bayesian framework along with MCMC sampling to infer the source parameters from observed contours. To this end, we design synthetic experiments of an oil spill from a fixed point source and conduct MCMC simulations to solve the Bayesian estimation problem of the source parameters. We demonstrate the robustness of the method to noise in the observations and missing data.

The paper is structured as follows. Section 2 describes the LPT model used in the experiments. Section 3 presents the proposed Bayesian approach and MCMC sampling technique. Section 4 describes two synthetic experiments designed to assess the relevance of the proposed method. Section 5 presents and discusses the numerical results. Final conclusions are presented in Section 6.

2. Model description

We use a 2D advection-diffusion LPT model for illustration purposes. The developed Bayesian machinery, however, is non-intrusive, and more complex models (or their surrogates) can be used instead. This allows to take advantage of already-existing codes (or software) developed to model application-specific processes. The model we employ simulates the transport and spread of oil on the sea surface without

accounting for weathering processes. For a spill released from a fixed point source, the model generates contours that delineate modeled oil slicks, from a computed concentration field.

2.1. The LPT model

The LPT model is a 2D random walk model governed by the stochastic differential equation

$$d\mathbf{x}_p(t) = \mathbf{u}(\mathbf{x}_p, t)dt + \sigma(\mathbf{x}_p, t)Z\sqrt{dt}, \quad (1)$$

where $\mathbf{x}_p(t)$ is the position (x_p, y_p) of the particle p at time t , $\mathbf{u}(\mathbf{x}_p, t)$ is the deterministic part of the flow field, corresponding to the mean Eulerian velocity field at the position $\mathbf{x}_p(t)$ and time t , and $\sigma(\mathbf{x}_p, t)Z\sqrt{dt}$ is a stochastic term that represents turbulent diffusion, with $\sigma(\mathbf{x}_p, t)$ a symmetric positive definite diffusion tensor characterizing a random motion

$$\sigma = \begin{pmatrix} \sqrt{2K_x} & 0 \\ 0 & \sqrt{2K_y} \end{pmatrix}, \quad (2)$$

K_x and K_y being the diagonal terms of the eddy diffusivity tensor in the two-dimensional Fokker-Planck equation (both set to $3 \text{ m}^2/\text{s}$ in this study), and Z is a zero-mean Gaussian noise with identity covariance matrix. Eq. (1) is solved for n_p particles using a first-order Euler integration scheme. The time step is set to 1 h, with 500 particles being released every time step, over the total duration of the release.

2.2. Concentration and contours

The concentration C of a contaminant, at a given location $\mathbf{x}$ and time t , is estimated from the Lagrangian particles by kernel density estimation (Yamada and Bunker, 1988), expressed as

$$C(\mathbf{x}, t) = \sum_{i=1}^{n_p} \gamma_i \phi_\sigma(\mathbf{x} - \mathbf{x}_i), \quad (3)$$

where γ_i is the mass carried by particle i , ϕ_σ a radial basis function, and n_p the total number of released particles. If we assume that all particles carry the same mass, we can write $\gamma_i = q/n_p$, where q is the total mass of the released contaminant. We choose the bivariate normal distribution (2D Gaussian) with variance σ^2 as the basis function, i.e.,

$$\phi_\sigma(\mathbf{x} - \mathbf{x}') = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{|\mathbf{x} - \mathbf{x}'|^2}{2\sigma^2}\right) \quad (4)$$

We set the standard deviation σ as two times the maximum value of the set defined by the distance between each particle and its closest neighbor at any given time (Beale and Majda, 1982a, 1982b, 1985; Knio and Ghoniem, 1990):

$$\sigma = 2 \times \max_i \left\{ \min_j (|\mathbf{x}_i - \mathbf{x}_j|) \right\}, \quad i, j = 1, \dots, n_p. \quad (5)$$

The contour or front Γ_τ of a contaminant is obtained from the concentration field as the contour line for a given value τ of the concentration. Given the density ρ of oil, this concentration threshold defines the thickness of oil at the edge of a slick, which is given by the ratio τ/ρ . Information about the minimum thickness of oil detected by SAR images (which is the thickness of oil at the edge of observed oil slicks) would help in this case to better tune this parameter. In the experiments, however, and for the sake of illustrating the proposed method, we set $\tau = 10^{-5} \text{ kg/m}^2$.

2.3. The flow field

A synthetic two-dimensional unsteady double-gyre flow is considered as the flow field that transports the oil in our experiments. The stream function of the flow is given by

$$\psi(x, y, t) = A \sin(\pi f(x, t)) \sin\left(\pi \frac{y}{L}\right), \quad (6)$$

where

$$f(x, t) = a(t)\left(\frac{x}{L}\right)^2 + b(t)\left(\frac{x}{L}\right), \quad (7)$$

$$a(t) = \varepsilon \sin(\omega t) \quad \text{and} \quad b(t) = 1 - 2\varepsilon \sin(\omega t), \quad (8)$$

over the spatial domain $[0, 2L] \times [0, L]$. The flow is stationary (steady) for $\varepsilon = 0$ and time-periodic for $\varepsilon > 0$. The x and y components of the velocity, u and v , are derived from the stream function as

$$u = -\frac{\partial \psi}{\partial y} = -\frac{\pi A}{L} \sin(\pi f(x, t)) \cos\left(\pi \frac{y}{L}\right), \quad (9)$$

$$v = \frac{\partial \psi}{\partial x} = \frac{\pi A}{L} \cos(\pi f(x, t)) \sin\left(\pi \frac{y}{L}\right) \left(L \frac{\partial f}{\partial x}\right) \quad (10)$$

In the experiments, we set $L = 200 \text{ km}$, $\omega = 2\pi/T$, where the period T of the flow is 7 days, $\varepsilon = 0.25$ and A such that $\pi A/L = 0.5 \text{ m/s}$. Snapshots of the flow field at $t = 0$, $t = T/3$ and $t = 2T/3$ are shown in Fig. 1.

3. Bayesian estimation of the source parameters

3.1. Bayes' rule

Let $\theta = \{\theta_i : i = 1, \dots, n_\theta\}$ and $\mathbf{d}$ denote the set of source parameters and the set of observations (or data), respectively. In a Bayesian framework, the parameters are modeled as random variables of prior probability distribution $\pi(\theta)$. The likelihood $\pi(\mathbf{d}|\theta)$ reflects how closely the model prediction matches $\mathbf{d}$, given θ as input. The estimation of θ is achieved by evaluating the posterior distribution, $\pi(\theta|\mathbf{d})$, which, based on Bayes' rule, can be obtained by combining the information about the prior and the likelihood (i.e., the "data") as

$$\pi(\theta|\mathbf{d}) \propto \pi(\mathbf{d}|\theta) \times \pi(\theta) \quad (11)$$

In general, the posterior distribution does not have a closed form, and therefore, sampling strategies, such as MCMC methods, are needed to estimate the posterior distribution. Statistics of the posterior (e.g., mode, mean, moments and credible intervals) can then be extracted, which help quantify the uncertainties in the parameters estimation. Any (point) estimator of θ can then be calculated, along with the associated uncertainties, based on the posterior distribution. Classical estimators are obtained based on classical loss functions that quantify the error made in evaluating θ by its estimate. These include, the posterior mean, $\bar{\theta} = \int \theta \pi(\theta|\mathbf{d}) d\theta$, which is based on the quadratic loss function, and the maximum a posteriori (MAP), $\hat{\theta} = \arg\max_{\theta} \pi(\theta|\mathbf{d})$, which is based on the zero-one loss function (Robert, 2007). In this work, we consider the MAP estimator.

3.2. The prior

Information about the inferred parameters, prior to the observations, is represented in the prior probability distribution, $\pi(\theta)$. As is usually the case, we assume that the parameters are a priori independent, i.e.,

$$\pi(\theta) = \prod_{i=1}^{n_\theta} \pi(\theta_i) \quad (12)$$

For the sake of generality, we further assume that information about the parameters is limited and choose non-informative priors, i.e., the marginal parameters follow uniform prior distributions over pre-defined bounds, i.e.,

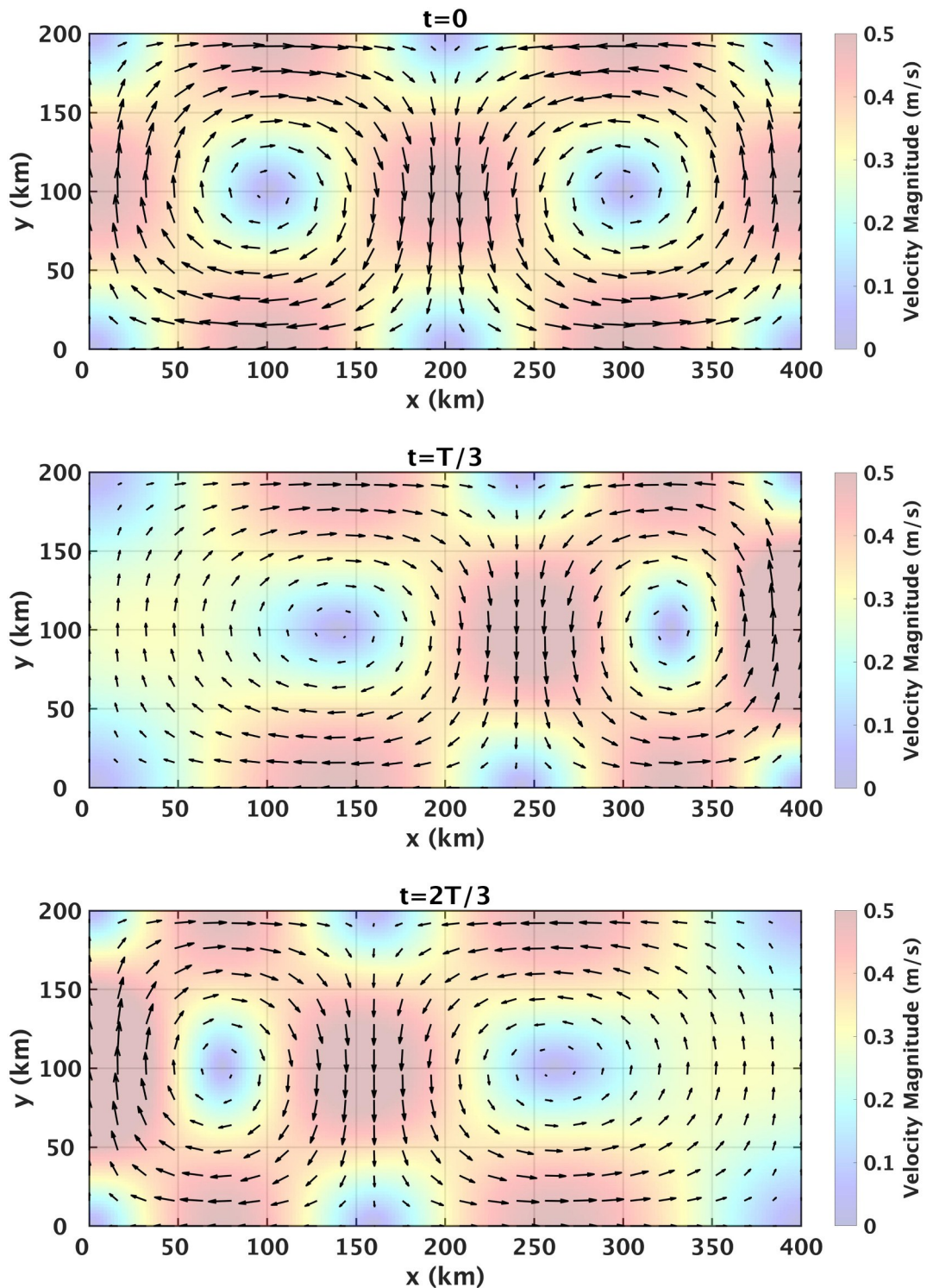


Fig. 1. Snapshots of the flow field used in the experiments, at $t = 0$, $t = T/3$, and $t = 2T/3$.

$$\pi(\theta_i) = \begin{cases} \frac{1}{b_{\theta_i} - a_{\theta_i}} & \text{for } \theta_i \in [a_{\theta_i}, b_{\theta_i}], \\ 0 & \text{otherwise.} \end{cases} \quad (13)$$

Other probability laws (e.g., Gaussian, truncated Gaussian) can be used to model the prior if further information about the parameters is available (Ait-El-Fquih et al., 2020).

3.3. The likelihood

As mentioned above, the likelihood $\pi(\mathbf{d}|\boldsymbol{\theta})$ expresses the probability that the (LPT) model output matches the observations, given $\boldsymbol{\theta}$ as a set of source parameters. This probability distribution is obtained by assessing the degree of mismatch (or dissimilarity) between the model prediction and the observations $\mathbf{d}$, based on a specified statistical relationship. In general, any metric (or dissimilarity measure) can be used to measure the “distance” between the observed and modeled contours, Γ and Γ_r .

We denote this metric by δ , i.e.,

$$\delta = \delta(\Gamma, \Gamma_r), \quad (14)$$

and adopt the exponential distribution as the likelihood function in our problem. Hence,

$$\begin{aligned} \pi(\Gamma|\theta) &= \text{Exp}(\delta; \beta), \\ &= \frac{1}{\beta} \exp\left(-\frac{\delta}{\beta}\right), \end{aligned} \quad (15)$$

where β is the scale parameter of the distribution. Comparing contours for shape matching has been widely used in the image recognition (and computer vision) community. Toyama and Blake (2002) used the exponential distribution to model the likelihood function in a shape matching problem. Gavrilu (2007) further incorporated additional degrees of freedom by choosing the gamma distribution as the likelihood (which generalizes the exponential distribution). In the atmospheric modeling community, a similar choice has been adopted in several studies (e.g., Jackson et al., 2004; Qian et al., 2018), in which the likelihood is proportional to the exponential of the negative of a scaled cost function. It is worth mentioning that, for a given β (the scale parameter of the distribution), maximizing the likelihood $\pi(\Gamma|\theta)$ is equivalent to minimizing the mismatch δ between Γ and Γ_r . This is similar to the equivalence between maximizing the Gaussian likelihood

$$\pi(d|\theta) \propto \frac{1}{\sigma_v} \exp\left(-\frac{|d - h(\theta)|^2}{2\sigma_v^2}\right), \quad (16)$$

and minimizing the squared error $|d - h(\theta)|^2$ in the classical measurement error model $d = h(\theta) + \nu$, where $h(\theta)$ is the model prediction and $\nu \sim \text{Normal}(0, \sigma_v^2)$.

For a set d of n_d independent contours we obtain

$$\begin{aligned} \pi(d|\theta) &= \prod_{i=1}^{n_d} \text{Exp}(\delta_i; \beta), \\ &= \frac{1}{\beta^{n_d}} \exp\left(-\frac{1}{\beta} \sum_{i=1}^{n_d} \delta_i\right), \end{aligned} \quad (17)$$

where δ_i is the distance from the i th observed contour Γ_i to the corresponding modeled contour (Γ_r, i) , i.e., $\delta_i = \delta(\Gamma_i, (\Gamma_r, i))$. The scale of the exponential distribution β (which is also its mean) represents, on average, the distance between the observed and true modeled contours, for different realizations of the observation error (which might be introduced by noise in SAR images affecting the observed contours). In the case where a decent number of observations (contours) is available, β can be estimated as a hyper-parameter along with the inferred source parameters. In the experiments presented here, we assume that β is known and set it to 100 km. This value was selected after assessing the distribution of the distance between predicted contours and samples of (pseudo-) observed contours generated as described in Section 4.

3.4. The dissimilarity measure

To measure the dissimilarity (i.e., δ) between observed and modeled contours, Γ and Γ_r , we employ the global Hausdorff distance (GHD). As in Tillier et al. (2013), we define this non-local metric as the L2 norm of the local Hausdorff distance (LHD), which is a 2D matrix:

$$\text{GHD} = \text{L2}(\text{LHD}) \quad (18)$$

The LHD was introduced by Baudrier et al. (2008) and is computed here as described in Zhang and Leeuwenburgh (2017). A binary image of a certain shape (an oil slick for example), limited by a contour, is defined as a matrix of zeros and ones, where the zeros are assigned to the matrix elements (or pixels) lying outside of the shape (or the contour), and the ones are assigned to the elements that are inside. Given two binary

images, I_A and I_B , of two shapes, A and B , respectively, the LHD between A and B is computed as

$$\begin{aligned} \text{LHD}(A, B) &= |I_A - I_B| \circ \max(D_A, D_B), \\ &= \max(|I_A - I_B| \circ D_A, |I_A - I_B| \circ D_B), \end{aligned}$$

where $\circ$ is the element-wise multiplication, and $\max$ denotes the element-wise maximum operator. D_A and D_B are distance maps of the shapes A and B . A distance map labels each pixel of the (binary) image with the (Euclidean) distance to the nearest pixel in the shape. Zhang and Leeuwenburgh (2017) also showed that the LHD can be constructed from two directed distance maps as

$$\begin{aligned} \text{LHD}(A, B) &= \max(I_B \circ D_A, I_A \circ D_B), \\ &= I_B \circ D_A + I_A \circ D_B, \end{aligned}$$

where $I_B \circ D_A$ represents the distance from A to B , and $I_A \circ D_B$ the distance from B to A (see Fig. 2).

3.5. Sampling the posterior

To sample the posterior distribution, we employ the Metropolis-Hastings (MH) algorithm (Gilks et al., 1995; Gelman et al., 2013), which is an MCMC technique in which samples collected from a distribution form a Markov chain. The posterior distribution can then be estimated from the collected samples by means of kernel density estimation (Hill, 1985). Each iteration $s = 1, \dots, n_s$ of the MH algorithm can be described by the following steps.

1. Suppose that the current state of the Markov chain is $\theta^{(s-1)} = \{\theta_i^{(s-1)} : i = 1, \dots, n_\theta\}$. Draw a candidate $\theta' = \{\theta'_i : i = 1, \dots, n_\theta\}$ from a Gaussian distribution centered at the current state

$$\theta' \sim \text{Normal}(\theta^{(s-1)}, \Sigma) \quad (19)$$

2. Evaluate the posterior probability distribution $\pi(\theta' | d)$, or more precisely, $\pi(d | \theta') \times \pi(\theta')$.
3. Calculate the acceptance probability

$$r = \min\left(1, \frac{\pi(\theta' | d)}{\pi(\theta^{(s-1)} | d)}\right) \quad (20)$$

4. Draw a sample α from a uniform distribution between 0 and 1, i.e., $\alpha \sim \text{Uniform}(0, 1)$.
5. The new state $\theta^{(s)}$ is chosen as

$$\theta^{(s)} = \begin{cases} \theta' & \text{if } \alpha < r, \\ \theta^{(s-1)} & \text{otherwise.} \end{cases} \quad (21)$$

To prevent drawing samples from outside of the prior bounds, we use reflection at the boundaries (Yang and Rodríguez, 2013; Thawornwattana et al., 2018). If a new value θ'_i is proposed, such that $\theta'_i > b_{\theta_i}$ ($\theta'_i < a_{\theta_i}$), we reflect that value back to the interval $[a_{\theta_i}, b_{\theta_i}]$ to obtain the proposed value $\theta'_i = b_{\theta_i} - (\theta'_i - b_{\theta_i})$ ($\theta'_i = a_{\theta_i} + (a_{\theta_i} - \theta'_i)$).

4. Numerical experiments design

Two numerical experiments were carried out to assess the effectiveness of the proposed framework. A total mass of $q_r = 100$ tons of oil is first released at time $t_r = 0$, over a period of $d_r = 3$ days, from a fixed point source located at $x_r = 100$ km and $y_r = 25$ km, and transported by the velocity field described in Section 2.3. In the first experiment (Exp. I), we assume that the oil slick is observed at time $t_o = 7$ days, i.e., after $\Delta t = t_o - t_r = 7$ days from the time of the initial release. The observed contour Γ of the oil slick is extracted from the simulated oil concentration, as described in Section 2.2. To mimic a realistic situation that includes the effect of speckle noise and generate a “noisy” contour, we

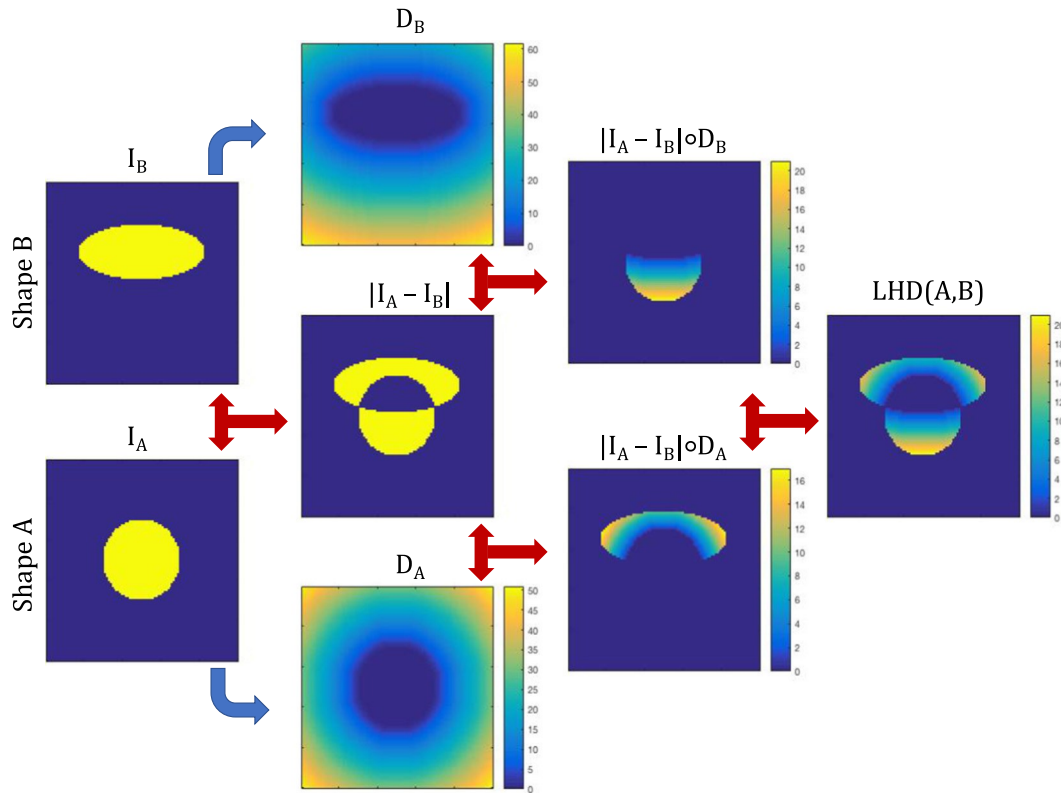


Fig. 2. Schematic illustrating the process of obtaining the local Hausdorff distance from the binary images of two shapes A and B.

perturb the computed concentration field with an additive zero-mean normally distributed noise. The noise has a standard deviation equal to the mean value of the concentration field within the unperturbed contour Γ_r , i.e., the subset bounded by Γ_r of the unperturbed concentration field. The observed contour Γ is then obtained as the contour line of the perturbed concentration field for the concentration threshold τ . Fig. 3a shows the “observed” contour Γ at day $t_o = 7$ (blue contour). The red curve represents the same contour before adding the noise (i.e., Γ_r). The red dot marks the location of the source of release. The blue vectors represent the flow currents at $t_r = 0$, the time of the initial release.

The second experiment (Exp. II) considers the same setup as in Exp. I, but assumes that the slick is observed at days $t_{o1} = 6$ and $t_{o2} = 8$. We further assume that the contours are partially observed within a specific

field of view. The goal of this experiment is to demonstrate how multiple observations (in time) of parts of a moving slick can be used in the proposed method. The observed contours are displayed in Fig. 3b along with the field of view represented by the red dashed rectangle. The parts of the contours outside that rectangle are assumed to be unobserved (missing data).

The proposed Bayesian inverse system is then implemented to infer the source parameters of the oil spill from the “observed” contours. We focus in particular on addressing the following questions. Where did the release occur?, when did it start prior to the time when observations became available?, for how long did the release last?, and how much oil was released overall? Using the observations generated in each experiment, we infer 5 parameters that would enable us to answer those

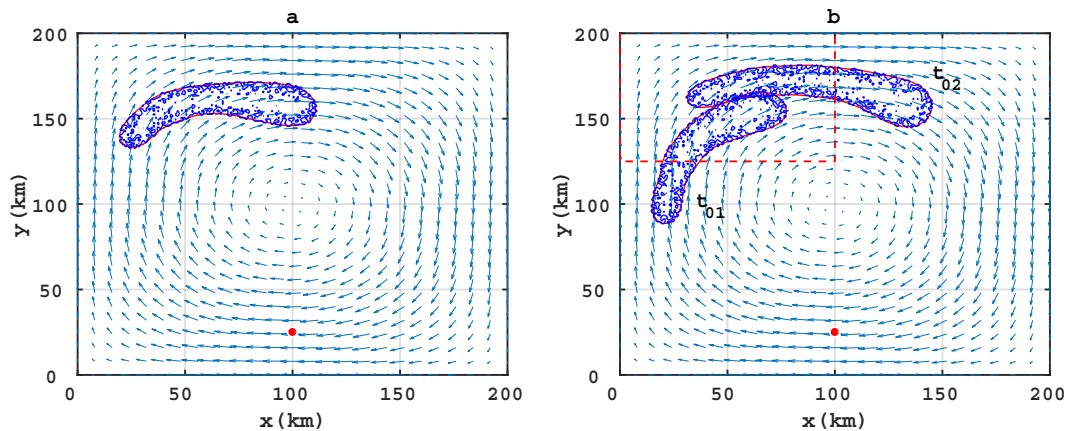


Fig. 3. Observations used in Exp. I (a) and Exp. II (b). The blue contours represent the edge of the slick observed at $t_o = 7$ days in Exp. I, and $t_{o1} = 6$ and $t_{o2} = 8$ days in Exp. II. The red contours are modeled contours (without noise), obtained based on the true (or reference) parameters. Observations in Exp. II are only available within a field of view shown by the red dashed rectangle in b. The red dot marks the location of the source of release. The blue vectors represent the state of the flow field at the time of release, $t_r = 0$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

questions; x_r , y_r , Δt , d_r and q_r . For consistency, we choose $t_{\text{ref}} = 7$ days in the inverse problem as our reference point in time from which the time t_r of release (which is to be estimated) can be measured in both experiments. In this case, the time of release would be measured with respect to t_{ref} and not to the time of observation (t_o in Exp. I and t_{o1} or t_{o2} in Exp. II), and hence, its reference value would be the same in both experiments, i.e., $\Delta t = t_{\text{ref}} - t_r = 7$ days. Table 1 shows the reference (or “true”) value and the bounds of the prior distribution, a_{θ_i} and b_{θ_i} , corresponding to each parameter θ_i .

5. Results and discussion

5.1. Exp. I

The posterior distribution $\pi(\theta|\mathbf{d})$ was sampled using the proposed MH algorithm. The number of MCMC iterations was set to 10,000. Fig. 4 shows the Markov chain obtained in the form of a trace plot. The blue line in each plot represents the samples of the parameter θ_i , whereas the red horizontal line marks the reference value of the corresponding parameter. The chain mixes well and explores the parameters' space. This can be noticed from the frequently oscillating pattern of the chain that samples around the reference values. The ratio of the newly accepted samples to their total number, or acceptance rate, is 35%.

For better representation of the posterior distribution, we ran 7 additional MCMC simulations to obtain a total of 80,000 samples (see Gilks et al. (1995) for a discussion about the choice of the number of chains). Each chain was initiated differently by sampling the initial state from the prior distribution. Note that the simulations can be run in parallel on a multi-core processor. For burn-in, we discarded the first 500 samples of each chain. The remaining samples of the 8 chains were then used as realizations of the source parameters to estimate the posterior distribution by means of kernel density estimation (Botev et al., 2010; Botev, 2016). This resulted in a 5-dimensional discrete probability distribution. For visualization, we extracted 2D slices of the posterior, in the (x_r, y_r) space, for different values of the parameters Δt , d_r and q_r . Fig. 5 shows those 2D maps (slices) for $\Delta t = 5, 6, 7, 8$ and 9 days, $d_r = 2, 3$ and 4 days and $q_r = 50, 100$ and 200 tons. The red dot in each plot marks the true (x_r, y_r) location of the source of release, whereas the black dot marks the point that has the highest posterior probability in each 2D map. The blue vectors in the background show the flow pattern at the time of release corresponding to each map, i.e., at $t_r = t_{\text{ref}} - \Delta t$. The blue contour represents the observation that is available at $t_o = 7$ days. The black curve is the output of the LPT model for a release from the black dot, for Δt , d_r and q_r that correspond to each plot. Since our prior distribution is uniform, the value of the posterior probability density function at a given point (x_k, y_k) in a 2D map, for which $\Delta t = \Delta t_k$, $d_r = d_k$ and $q_r = q_k$, is proportional to that of the likelihood, and reflects the similarity between two contours: the observed contour (shown in blue), and a modeled contour resulting from a hypothetical release at (x_k, y_k) , simulated at Δt_k days prior to t_{ref} , and that lasted for d_k days, during which q_k tons of oil were spilled.

For fixed values of d_r and q_r (looking at a single column of plots in Fig. 5, for a constant d_r), the highest-probability point of a 2D map (i.e., the black dot) gets closer to the observation, along the streamlines of the flow, as Δt decreases. In other words, if Δt is small, there are higher

chances that the source would be closer to the observation, along the flow, than if Δt is large. This is related to the distance that the slick covers within Δt , given the form of the ocean currents. For fixed values of Δt and q_r , we notice that the larger the duration d_r of the spill, the closer the black dot is to the observed slick along the flow. If the duration d_r is larger, the resulting slick is expected to be larger in size. Therefore, a better match between the modeled and observed slicks, in terms of the Hausdorff distance, is obtained if the slicks are ‘centered.’ This happens if the source is closer to the observation. An analogous interpretation can be made for the case when d_r is small. Varying q_r , while d_r and Δt are fixed, does not seem to have an important effect on the position of the black dot. In fact, changes in q_r only affect the size of the slick, if all other parameters remain the same. The more oil is spilled (larger q_r), the larger the size of the resulting slick, and vice-versa. Varying q_r has a similar effect as varying the concentration threshold τ of the contour (as well as the thickness of oil at the edge of the slick). Increasing q_r for instance, for a given concentration threshold, would result in a slick similar in size to a one obtained for the same quantity q_r but for a smaller concentration threshold τ . Therefore, a strong negative monotonic relationship seems to exist between q_r and the concentration threshold used to define a contour. This strong relationship leads to an undetermined problem in the sense that it is difficult to infer both q_r and the concentration threshold together, since different combinations of the two parameters yield the same result.

A visual inspection of the different 2D maps allows us to determine the point that has the highest posterior probability (the MAP). This point, which we denote by A, is indicated as a black dot in the plot corresponding to $d_r = 3$ days, $\Delta t = 7$ days and $q_r = 100$ tons. In the same plot, the red dot, which represents the true parameters (see Table 1), lies in a high-probability region (close to point A), indicating that the method has successfully provided a good estimate of the solution of the inverse problem. Another interesting observation is the good match between the blue and black contours in the plot at $d_r = 3$ days, $\Delta t = 6$ days and $q_r = 100$ tons. This match is reflected by the high posterior probability at the black dot in the same plot (which we will call B). The high probability of that point, along with the close match between the two contours, suggests that point B would be also a good estimate of the solution. A more informative prior in this case would have helped to set the true parameters apart from other probable parameters. For instance, the estimated solution could have been improved if the prior had assigned more weight on earlier times (i.e., larger Δt 's). At this point, two sets of parameters associated to points A and B appear to be good point estimates. A relevant question is whether an observation at a previous time (before $t_o = 7$ days) would have helped to provide a better estimation by increasing the difference in probability between point A (which is close to the true parameters) and point B, and present A as a unique point estimate. To answer that, we plot the contours resulting from the true parameters and from point B (for which $\Delta t = 6$ days, $d_r = 3$ days and $q_r = 100$ tons) at different times before $t_o = 7$ days. Those contours are shown in Fig. 6, one, two, and three days prior to t_o (or t_{ref}). As can be seen, the contours have a close match, even 3 days prior to t_o , when the release from point B had just stopped. Therefore, observations at a previous time would still result in a high posterior probability around point B.

A factor that affects the similarity between slicks resulting from different scenarios is the variability in the ocean currents. This variability plays a role in the ‘geometric’ transformations that a slick undergoes, e.g., stretching where there is divergence in the flow, and compression where there is convergence. The higher the variability, the less is the potential similarity between two different slicks. To illustrate this point, we conducted 3 MCMC experiments (not shown) where we changed the variability in the currents by varying the period of the flow (the double-gyre). In the first, the period was halved (i.e., $T = 3.5$ days), in the second, it was doubled (i.e., $T = 14$ days), and in the third, the flow was steady ($\varepsilon = 0$, see Section 2.3). Indeed, the results of the experiment with high variability in the currents ($T = 3.5$ days) showed a

Table 1

True (reference) values and prior bounds for each inferred parameter θ_i .

Inferred parameter θ_i	True value	Prior bounds	
		a_{θ_i}	b_{θ_i}
x_r (km)	100	0	200
y_r (km)	25	0	100
Δt (days)	7	5	9
d_r (days)	3	1	5
q_r (tons)	100	0	500

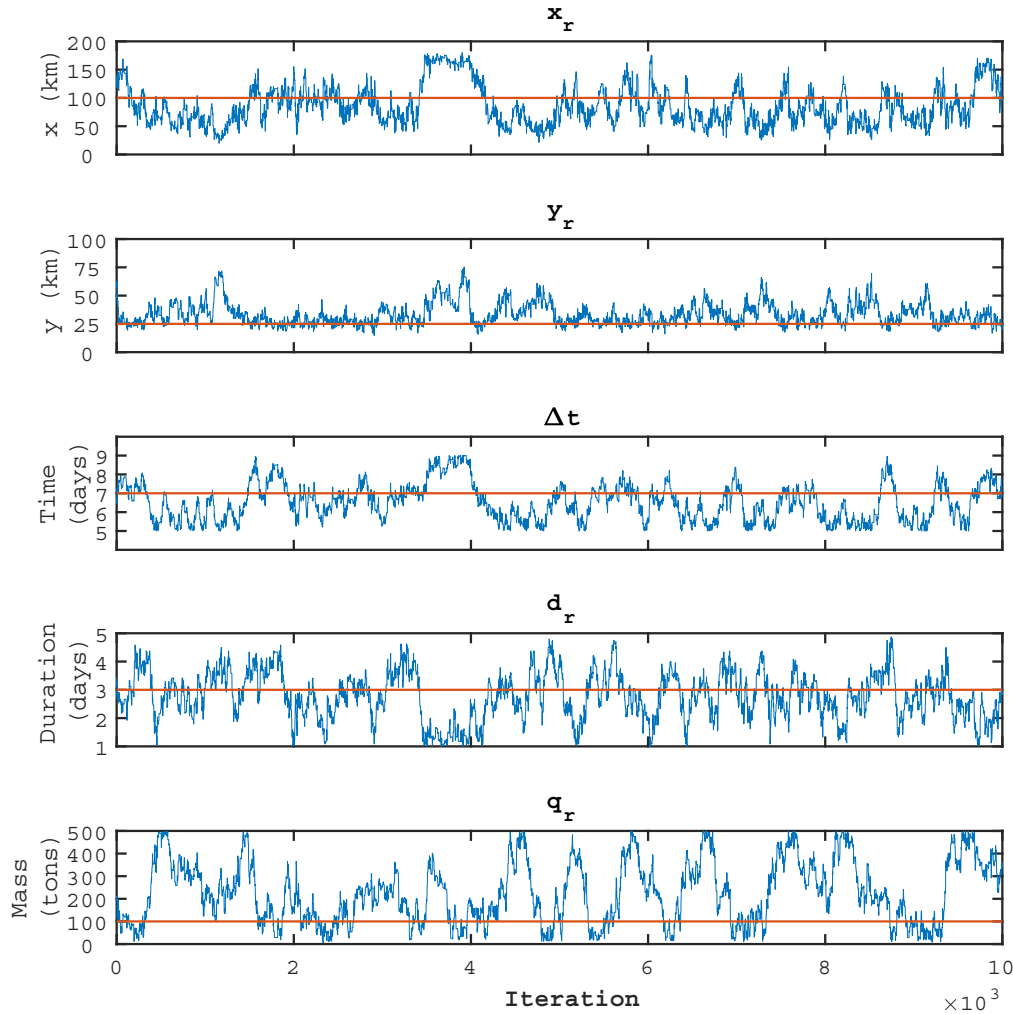


Fig. 4. Markov chain of an MCMC simulation of Exp. I consisting of 10,000 iterations, shown as trace plots of the five parameters x_r , y_r , Δt , d_r and q_r . The red horizontal line in each plot represents the value of the corresponding true (reference) parameter. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

high posterior probability only around the true parameters, whereas in the other experiments where the variability is low, the posterior probability was flatter and high probability regions were seen across different 2D maps. It is worth noting that, although high variability in the currents might lead to improved estimates, it is usually associated with high uncertainties in the flow, which could lead to higher uncertainty in the final estimates.

The relationships between the five parameters are further explored in a corner plot. This chart, shown in Fig. 7, allows us to visualize the relationship between any pair of variables, in the form of joint probability distributions, as well as the marginal distributions of the variables. Those distributions are obtained from the MCMC samples by means of kernel density estimation (Hill, 1985). The red lines in each plot mark the values of the corresponding true parameters. The red dashed lines represent the trend of the samples in each plot; the red dashed curved lines are second-order polynomial fits, whereas the red dashed straight lines are linear fits. The relationship between x_r and y_r is a nonlinear u-shaped relationship; y_r goes above 25 km whenever x_r goes above or below 100 km. This behavior is reflected by the form of the flow (the gyre) in the domain where x_r and y_r are sampled. A positive monotonic relationship can be observed between x_r and Δt . y_r and Δt exhibit a nonlinear relationship in which y_r tends to go above 25 km whenever Δt goes above or below 7 days. This behavior, along with the positive relationship between x_r and Δt , move the inferred source upstream

closer to the observation, as Δt becomes smaller. A negative monotonic relationship exists between x_r and d_r , and from the joint distributions between y_r and d_r , and Δt and d_r , one can notice positive relationships between the two pairs of variables (although with different slopes of the trend lines and a weak relationship between Δt and d_r). This indicates that a longer duration of release is expected whenever the source moves upstream closer to the observation (smaller x_r) and to the center of the gyre (larger y_r), or whenever the release occurs further back in time (larger Δt), and vice versa. q_r does not seem to have any effect on x_r and y_r as can be seen from the vertical trend line in both (x_r, q_r) and (y_r, q_r) plots. This is consistent with what we have observed previously in the posterior maps, where varying q_r did not seem to have any effect on the position of the black dot in Fig. 5. A slightly negative relationship (although weak) can be noticed between q_r and Δt , and q_r and d_r , which translates into a smaller quantity of oil being released, if the release occurs further back in time, or if it lasts longer, and vice versa. The plots on the diagonal show the marginal distributions. All reference parameters have relatively high marginal probabilities.

5.2. Exp. II

In this experiment, observations of the oil contour are partially available at two different instances, $t_{o1} = 6$ and $t_{o2} = 8$ days. As in Exp. I, we ran 8 MCMC simulations of 10,000 iterations each to obtain a total of

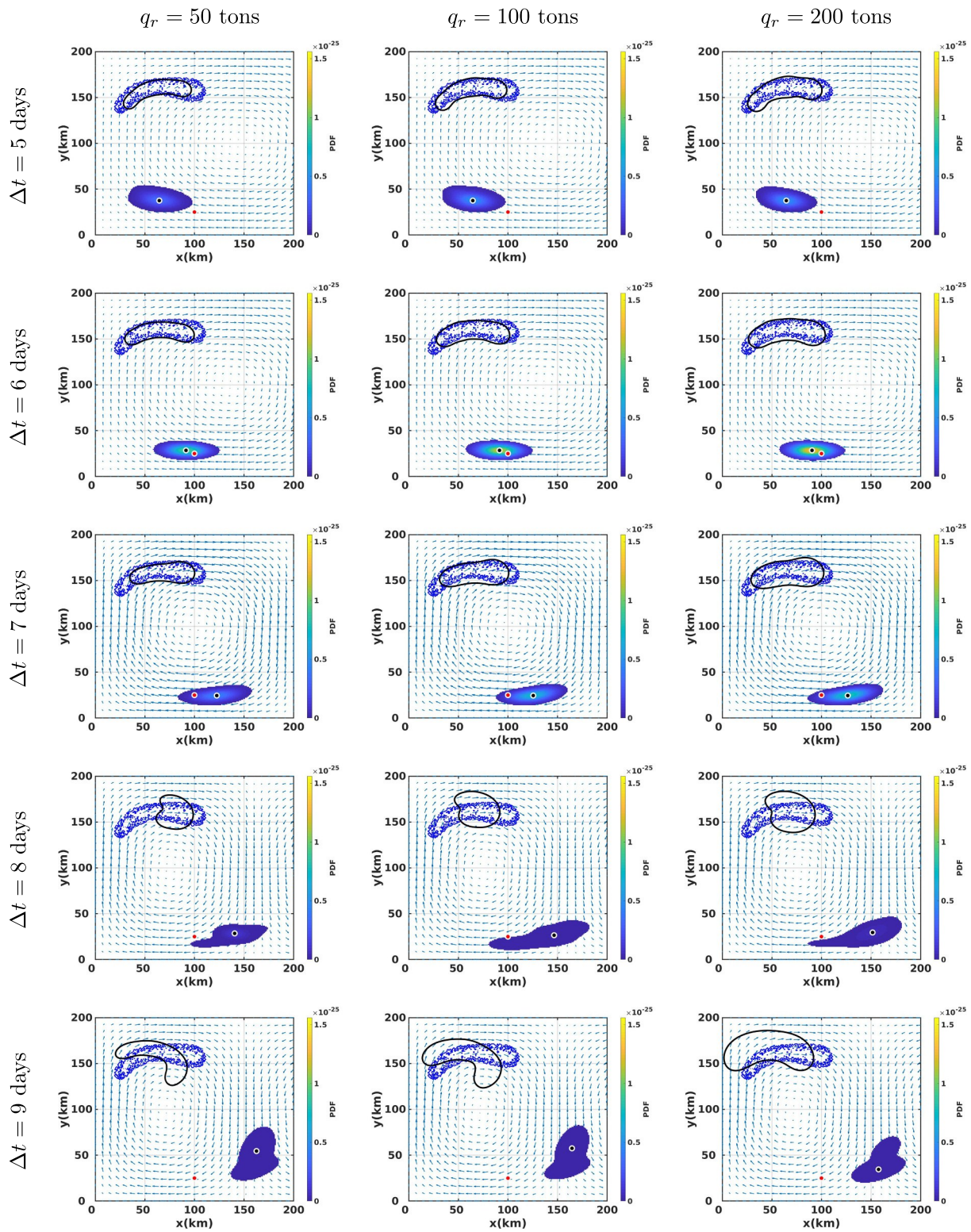


Fig. 5. (x_r, y_r) maps (slices) of the 5D posterior probability distribution of Exp. I, for $d_r = 2$ days, $\Delta t = 5, 6, 7, 8$ and 9 days, and $q_r = 50, 100$ and 200 tons. The red dot in each plot marks the true values of (x_r, y_r) . The black dot marks the (x_r, y_r) location that has the highest posterior probability in each 2D map. The blue contour represents the observation, whereas the black contour is the model output of a hypothetical release from the black dot, for values of d_r , Δt and q_r corresponding to each plot. The blue vectors show the currents at the time of release corresponding to each map, that is $t_0 - \Delta t$.

Continued for $d_r = 3$ days.

Continued for $d_r = 4$ days. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

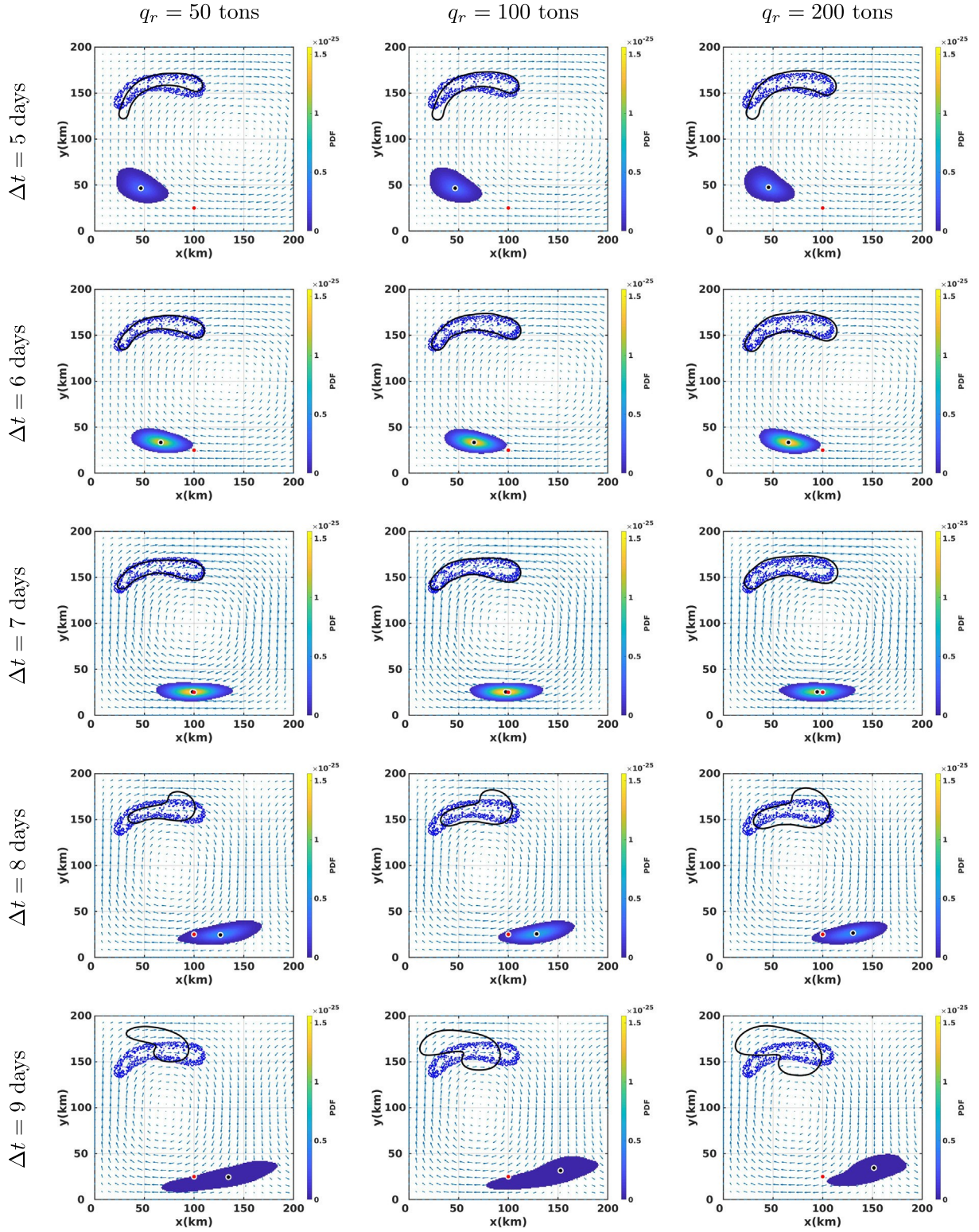


Fig. 5. (continued).

80,000 samples, with an acceptance rate of 29%. The first 500 samples of each chain were discarded for burn-in. Fig. 8 shows 2D probability maps, in (x_r, y_r) , of the posterior distribution, for $d_r = 2, 3$ and 4 days, $\Delta t = 5, 6, 7, 8$ and 9 days, and $q_r = 50, 100$ and 200 tons.

Similar conclusions to those made in Exp. I can be made here, in terms of how variations in d_r , Δt and q_r affect the location of the black dot (the point that has the highest posterior probability in each 2D map).

The point corresponding to the highest estimated posterior probability in all maps (the MAP) is the black dot in the plot at $d_r = 3$ days, $\Delta t = 7$ days and $q_r = 100$ tons. This point coincides with the red dot in the same map, which is associated with the reference parameters. This suggests that the proposed method provided a good estimate of the solution, given the available partial observations. Note, however, that the field of view of the observations captures the head of the slick at t_{o1} , and its tail

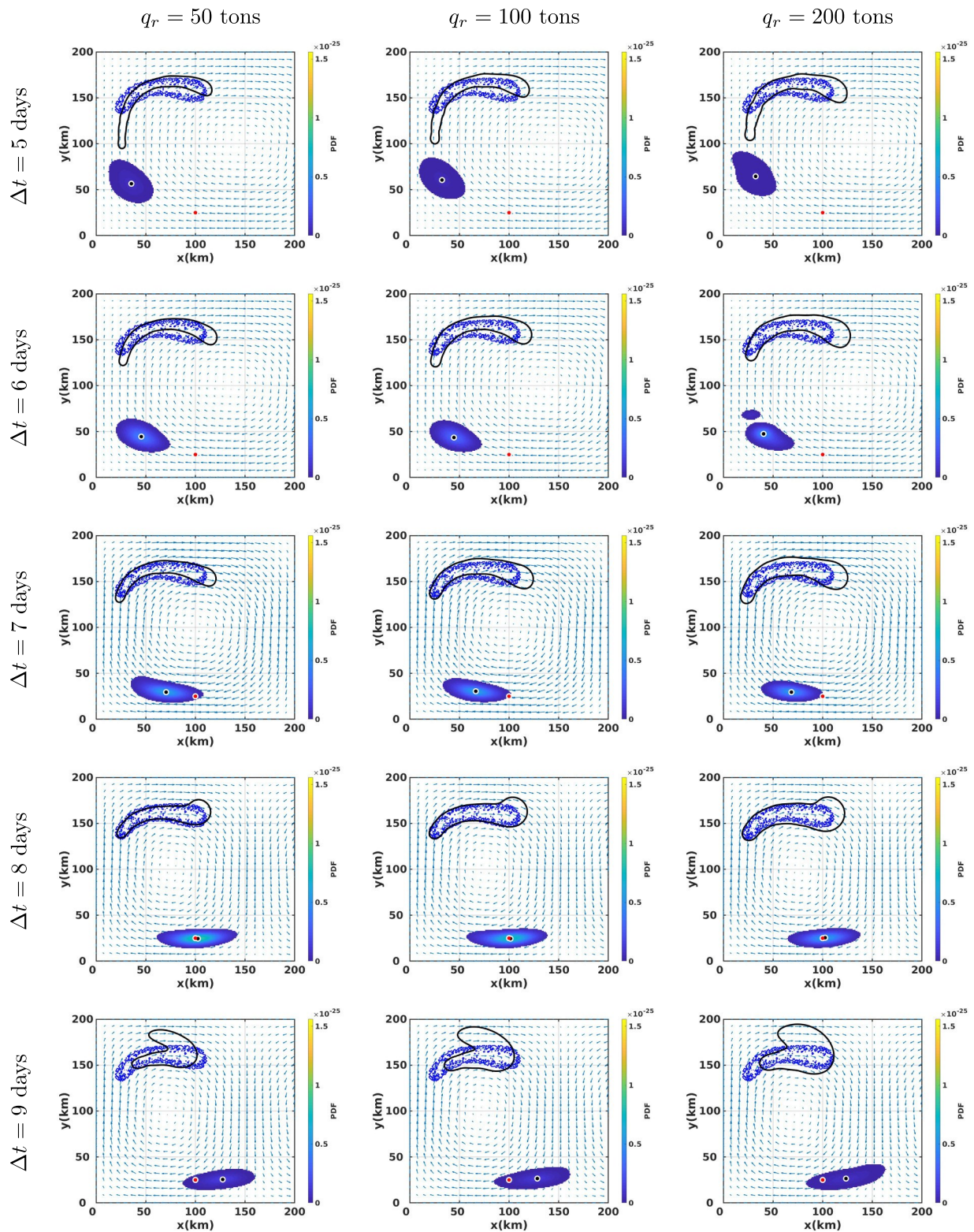


Fig. 5. (continued).

at t_{02} . A single partial observation, or multiple observations of the same part of a slick might result in an estimate with higher uncertainties, i.e., a flatter posterior distribution. This is because many hypothetical sources may result in slicks that match part of the observation, but not the other, especially in flows with low variability. It is worth mentioning that a good estimation of the source parameters (as in this experiment) would help to reconstruct the missing parts of an observation (as shown

in the plot containing the point with the highest posterior probability), and thus, allows to make better forecasts by initiating the LPT model from the reconstructed observations.

6. Conclusion

Marine ecosystems and coastal environments face a serious threat

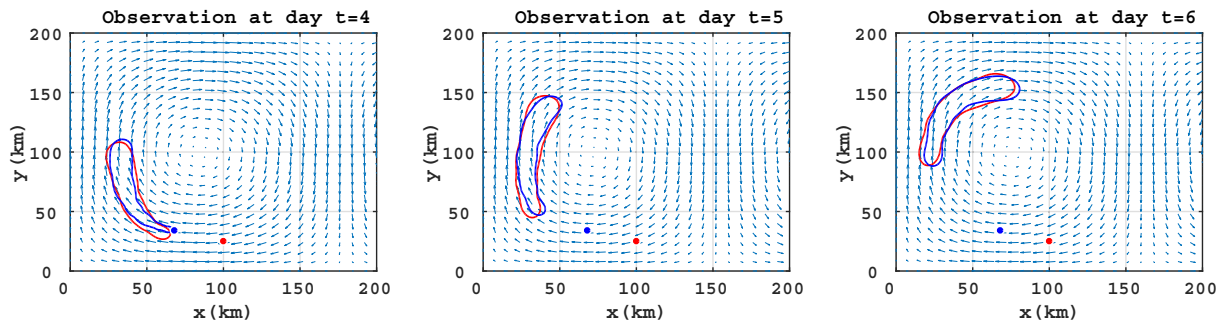


Fig. 6. Contours resulting from two different releases observed at days $t=4, 5$ and 6 (or $3, 2$ and 1 days prior to t_0 , respectively). The red contour corresponds to a slick that originated from the red dot, which represents the reference (true) source parameters. The blue contour corresponds to a release from point B, which is marked by the blue dot, for $\Delta t = 6$ days, $d_r = 3$ days and $q_r = 100$ tons. The blue vectors represent the currents at the time when the contours were observed. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

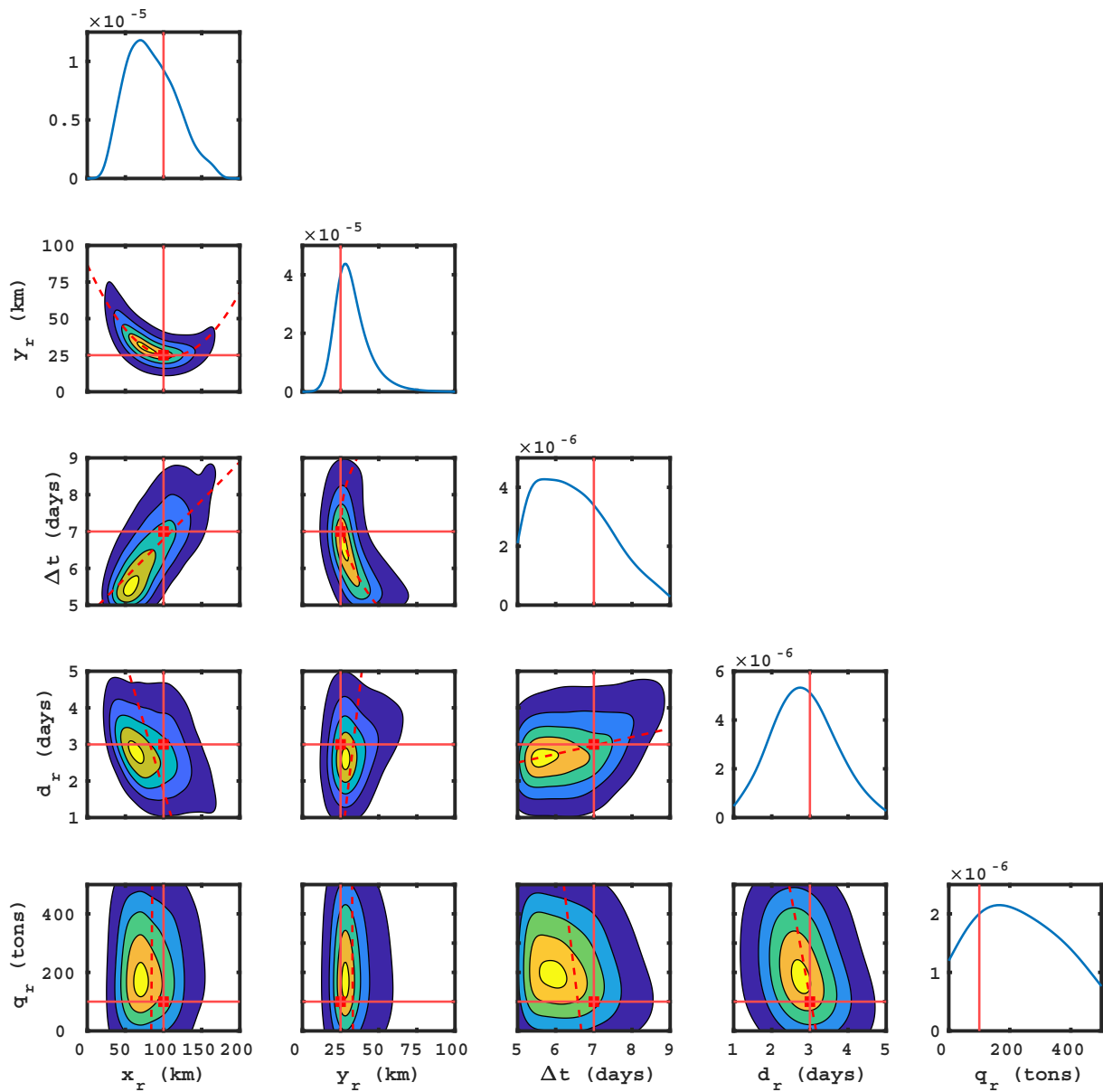


Fig. 7. Corner plot of the five parameters $x_r, y_r, \Delta t, d_r$ and q_r , formed by plots of the joint probability distributions of each pair of the parameters and plots of the marginal distributions of the single parameters. The red lines represent the values of the true parameters corresponding to each plot. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

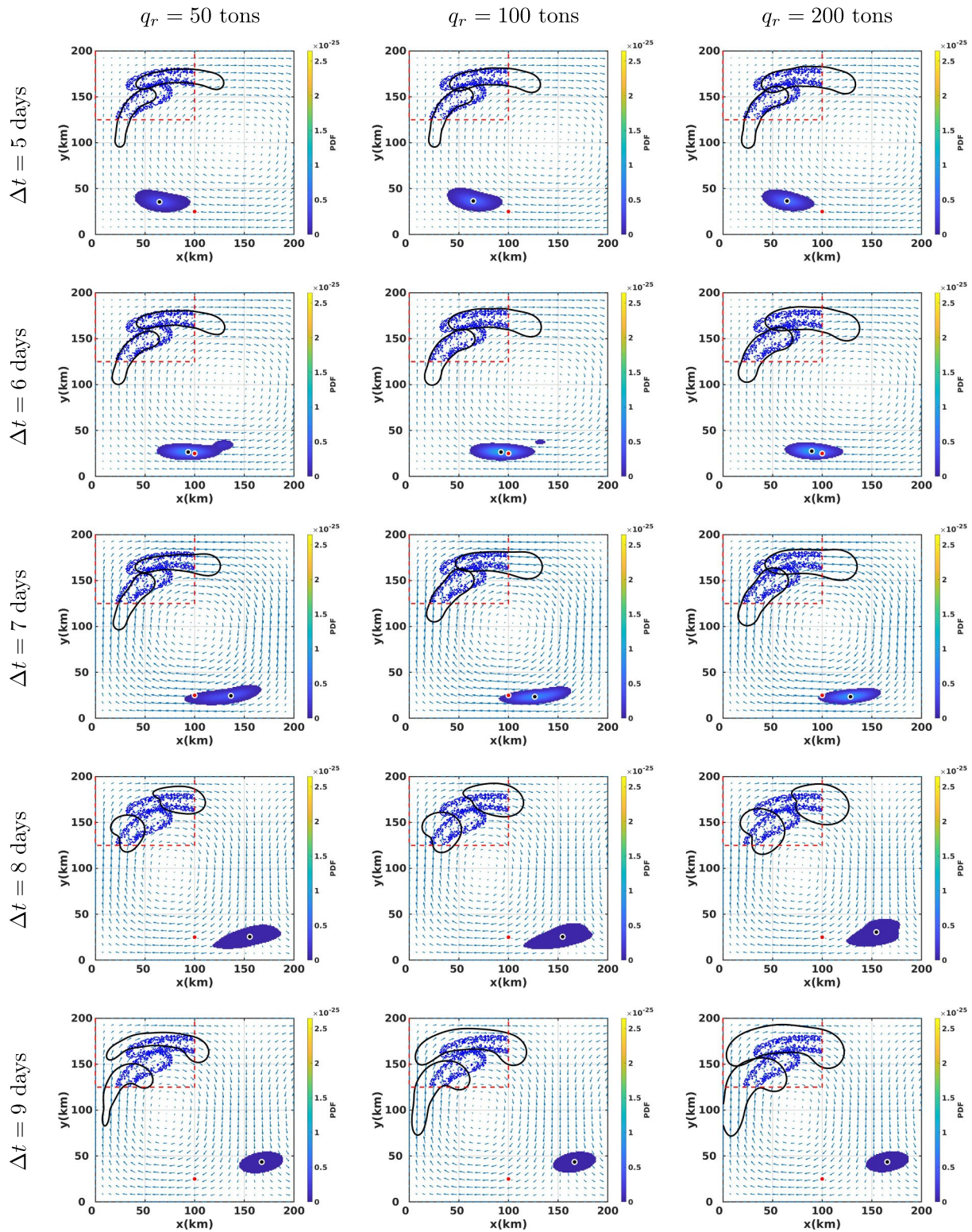


Fig. 8. (x_r, y_r) maps (slices) of the 5D posterior probability distribution of Exp. II, for $d_r = 2$ days, $\Delta t = 5, 6, 7, 8$ and 9 days, and $q_r = 50, 100$ and 200 tons. The red dot in each plot marks the true values of (x_r, y_r) . The black dot marks the (x_r, y_r) location that has the highest posterior probability in each 2D map. The blue curves represent the observations, whereas the black contours are the model output of a hypothetical release from the black dot, for values of d_r , Δt and q_r corresponding to each plot, shown at t_{o1} and t_{o2} . The blue vectors show the currents at the time of release corresponding to each map, that is $t_{ref} - \Delta t$.

Continued for $d_r = 3$ days.

Continued for $d_r = 4$ days. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

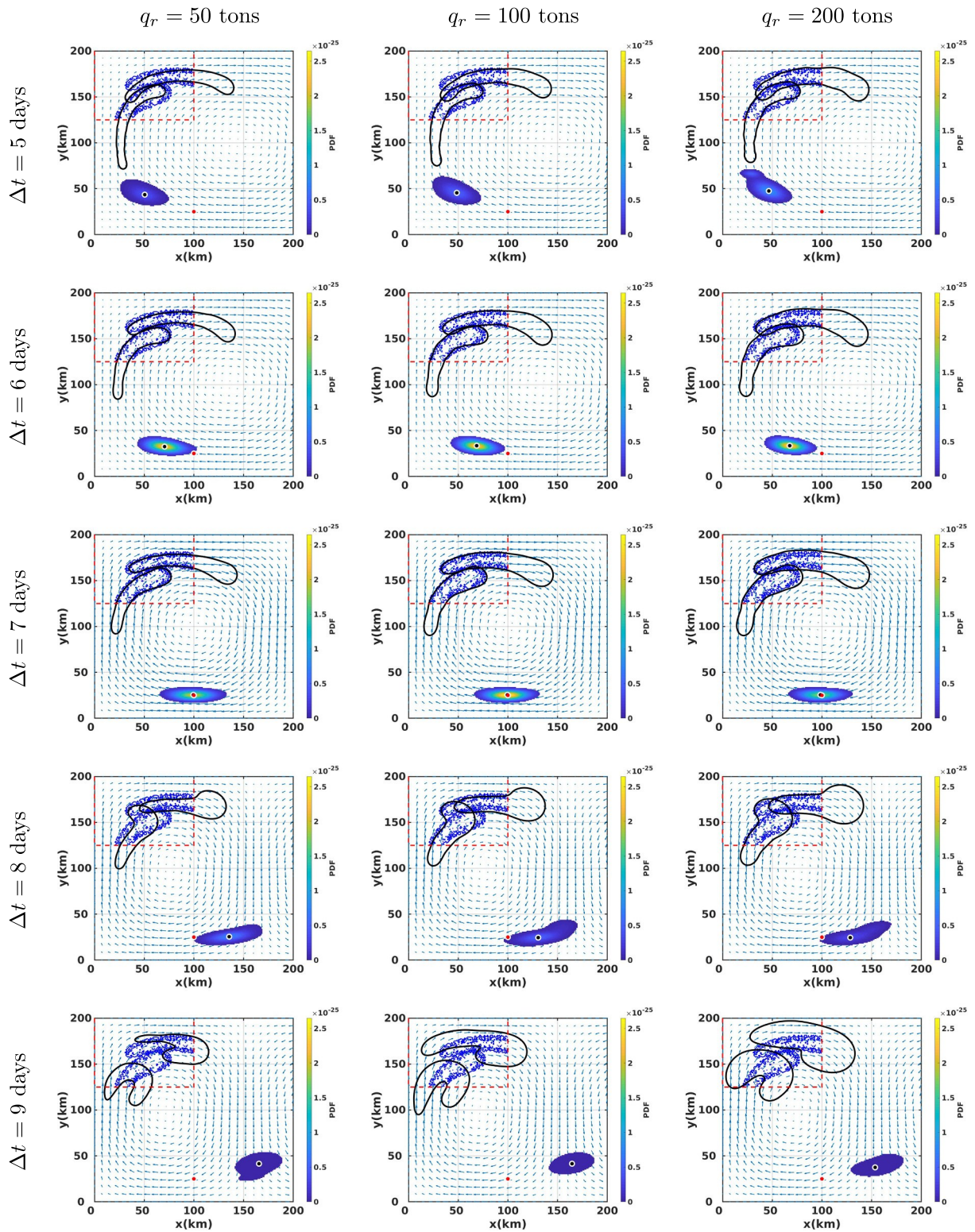


Fig. 8. (continued).

due to oil spills. The ability to determine the source of pollution from available observations, such as satellite images of oil slicks, may help to limit and manage such polluting events. We presented in this paper a generic Bayesian framework, in which LPT models can be used, to infer source parameters from images of oil contours. The posterior probability resulting from this approach reflects the uncertainties in the estimate, and thus a level of confidence in the inferred parameters. Two synthetic

experiments were designed to assess the relevance of the method. In the first experiment, a single observation of an oil slick was available. In the second one, the slick was partially observed at two different times. The method proved to be successful at providing good estimates of the source parameters, which consisted of the x and y coordinates of the source of release, the time when the release happened, the duration of the release, and the total quantity of pollutant released. We investigated

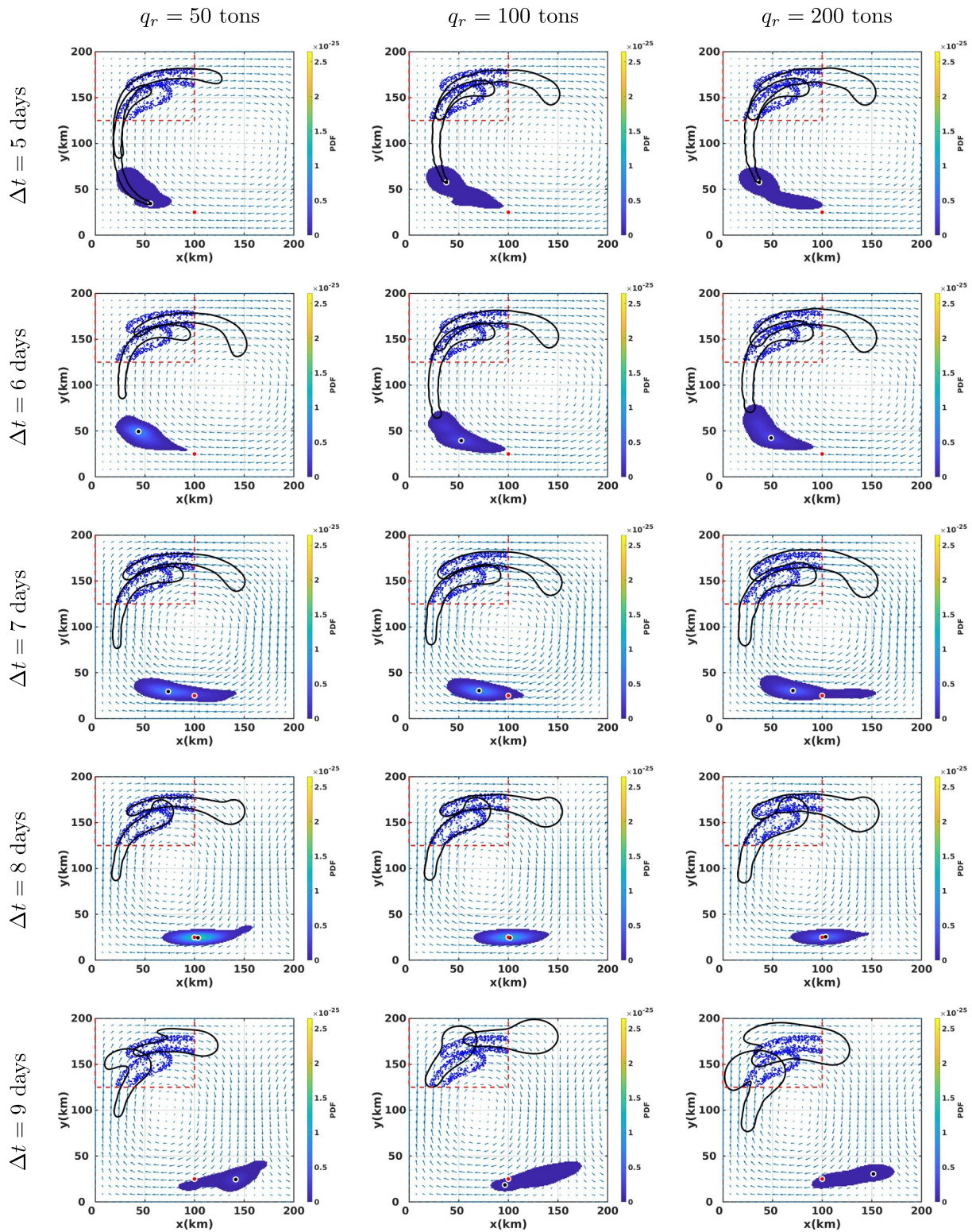


Fig. 8. (continued).

in the first experiment the sensitivity of the estimation (through the posterior distribution) to the variability in the flow field. Uncertainties in the estimation are higher in the case of near-steady flows or flows that exhibit low variability. More informative priors might help in this case to improve the estimation and reduce the uncertainties. We also showed in the second experiment how inference from observations of parts of a slick helps to reconstruct the missing parts of the observations. Initiating

oil spill models from the reconstructed observations may thus allow making better forecasts of the fate and transport of the oil slick.

This study used a simple LPT model in the numerical experiments to demonstrate the proposed approach. Using more-computationally-demanding models may not be feasible within MCMC. One may consider building a surrogate of the LPT model, using, for instance, a polynomial chaos expansion (e.g., [Marzouk et al., 2007](#); [Marzouk and](#)

Najm, 2009; Sraj et al., 2013; Giraldi et al., 2017; Sraj et al., 2017; Navarro et al., 2018). This functional representation, which substitutes the original model within MCMC, allows for fast sampling of the posterior distribution. This work can be extended by 1) further developing the method to allow for the estimation of moving source parameters, and 2) testing the method in real events of oil spills. These tasks will be the focus of our future work.

CRedit authorship contribution statement

Samah El Mohtar: Conceptualization, Methodology, Software, Investigation, Writing – original draft, Visualization. **Boujemaa Ait-El-Fquih:** Methodology, Investigation, Writing – review & editing. **Omar Knio:** Conceptualization, Methodology, Resources, Writing – review & editing, Supervision. **Issam Lakkis:** Conceptualization, Writing – review & editing. **Ibrahim Hoteit:** Conceptualization, Methodology, Resources, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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